

TECHNICAL DESIGN DOCUMENT

PRAMANA

प्रमाण | *the means by which a thing is known to be true*

An Over-Determined Digital Twin for Aero Piston Engines in MALE UAVs

Problem Statement **SIH26054** | Defence Research and Development Organisation
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DESIGN THESIS

Conventional engine health monitoring compares a measurement to a limit. A digital twin compares a measurement to a physics-based prediction. **PRA-MANA compares every measurement to every other measurement.** An internal combustion engine is a massively over-determined system: the same physical quantity is recoverable along several independent measurement paths, and health is the pattern of disagreement between them. Every capability in this document — virtual sensing, sensor-fault discrimination, fault isolation, cross-engine transfer, and validation without ground truth — is a consequence of that single structural fact.

Document prepared for internal team use and for evaluation submission. All engine limits are quoted from published type-certificate data; all platform figures from published DRDO and open sources. Performance figures generated from the simulation environment described in §4 are identified as such throughout.

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1 Executive Summary

PRAMANA is a real-time digital twin for the turbocharged aero piston engines used in Medium Altitude Long Endurance unmanned aircraft. It ingests engine telemetry over CAN, runs a physics-based engine model in lockstep with the live data, and reports component-level health, fault diagnosis, remaining useful life, and a mission-completion probability with an associated recommended operating envelope.

The system differs from conventional health monitoring in three structural ways.

It is over-determined by construction. Air mass flow, brake power and the engine energy balance are each computed along multiple independent physical paths. In a healthy engine with honest sensors these agree; each fault mode violates a specific, derivable subset of the consistency relations. Fault isolation is therefore a consequence of physics rather than a hand-authored lookup table, and the twin can be validated against itself in the complete absence of run-to-failure ground truth — which is the central practical obstacle in this domain.

It separates sensor faults from engine faults. A degrading component perturbs several coupled parity relations in a physically consistent pattern. A drifting transducer perturbs only the relations that contain it. The distinction is recoverable from the structure of the residual vector, and where it is ambiguous the system resolves it actively, by commanding a small diagnostic perturbation and testing the *gain* of the response rather than its level.

It acts rather than reports. The problem statement asks for mission reliability *enhancement*. PRAMANA propagates the identified degradation state forward over the remaining mission profile, returns a probability of completion, and solves for the power and boost schedule that maximises mission value subject to a reliability constraint — a renegotiated operating envelope rather than an alarm.

WHAT IS AND IS NOT CLAIMED

None of the individual techniques used here are novel. Mean-value engine modelling, parity-space fault detection, joint state-parameter estimation, active fault diagnosis and life-extending control are all established methods with substantial literature. The contribution claimed is their coherent assembly into a working system for a UAV piston engine, together with an explicit answer to the validation problem that this application domain has never had a good answer for.

2 The Problem

2.1 Problem statement SIH26054

DRDO's statement is titled "*AI-Enabled Real-Time Digital Twin System for Health Monitoring, Fault Prediction and Mission Reliability Enhancement of Aero Piston Engines used in MALE UAVs*". Its background section is unusually direct about what is wrong with the status quo:

"Conventional engine monitoring systems used in UAVs are primarily threshold-based and reactive in nature." Existing approaches, it continues, cannot estimate remaining useful life, predict degradation trends, or simulate mission-wise engine behaviour.

The expected solution is specified in six lettered components: (A) a modular digital twin core synchronised to live data; (B) continuous health monitoring across RPM, cylinder head and exhaust gas temperatures, oil metrics, fuel flow, vibration signature, battery and alternator health, and injection timing; (C) fault detection covering misfire, injector abnormality, lubrication faults, *sensor drift*, combustion instability, overheating trends and vibration patterns; (D) an AI/ML layer providing anomaly detection, RUL estimation and maintenance recommendation; (E) simulation and replay including high-altitude and endurance scenarios; and (F) a visualisation dashboard. CAN bus interfaces, edge computing and cloud analytics are named explicitly, and a deployment roadmap is a listed deliverable.

Two items in that list deserve emphasis because they are easy to overlook. *Sensor drift* appears as a fault class alongside genuine engine faults — the statement is asking for a system that can tell them apart. And the split between edge and cloud is not stylistic: it is a consequence of datalink bandwidth, discussed in §12.

2.2 Operational context

The relevant DRDO powerplant is the 180 hp aero-diesel developed by the Vehicles Research and Development Establishment, which holds rated power to 11 000 ft, runs an indigenously developed FADEC, has accumulated over 1100 hours on test profiles, and has been trialled at Leh and Changla to 17 664 ft. The TAPAS BH-201 platform is specified with two such engines at 220 hp in the production configuration, against 100 hp prototype units; it has demonstrated a 28 000 ft ceiling and 18 h endurance against targets of 30 000 ft and 24+ hours.

Three properties of this operating regime shape the design. The aircraft is **twin-engined**, which supplies a free reference channel (§8). Missions are **long and repetitive**, dominated by loiter at a fixed altitude and power setting, which supplies naturally repeated samples at an identical operating point. And the aircraft operates **far above the engine's critical altitude**, where turbocharger condition dominates available power and where a degradation invisible at sea level becomes mission-limiting.

2.3 Why threshold monitoring fails

A limit-based monitor answers one question: has this channel exceeded a fixed number? That formulation has three failure modes, all of which appear in normal UAV operation.

- **It is late by construction.** A limit is placed where damage begins, so an alarm is by definition the announcement that damage has begun. A parameter drifting steadily toward a limit carries the same information hours earlier, but only relative to what it *should* be at that operating point — which a fixed threshold does not know.
- **It cannot use context.** Cylinder head temperature of 128 °C is benign at maximum continu-

ous power on a hot day at low level and is a serious anomaly at cruise power at 20 000 ft. A single number cannot encode that, so the limit is set for the worst case and loses sensitivity everywhere else.

- **It cannot doubt its own input.** A limit exceedance and a failed transducer are indistinguishable to a threshold. In an unmanned aircraft with no crew to cross-check the instrument, this produces mission aborts caused by instrumentation rather than by engines.

Notably, the Rotax 914 type certificate — the most completely documented engine in this class — publishes no exhaust gas temperature limit at all. EGT on these engines is a trend parameter, not a limit parameter. A threshold system has no way to use it. This is precisely the information a twin exists to exploit.

3 Design Thesis: Over-Determination

3.1 The structural fact

An instrumented internal combustion engine carries more measurements than it has independent states. This redundancy is normally treated as a nuisance — more channels to display, more limits to set. It is in fact the most valuable property the system has, because it means the measurements can be checked *against each other* without any external reference.

Consider air mass flow. It is not directly measured on most aero piston installations, yet it can be inferred four different ways from instrumentation that is already present: from the speed–density relation, from the compressor operating point, from the fuel flow and measured excess-air ratio, and from a flow restriction in the intake. Each inference uses a different subset of the sensors and a different piece of physics. In a healthy engine with honest transducers, all four return the same number.

THE CORE CLAIM

Health is not the distance of a measurement from a limit. Health is the **degree of mutual consistency** among redundant estimates of the same physical quantity. Degradation and instrumentation faults each destroy that consistency, but they destroy it in different, derivable patterns — and the pattern identifies the cause.

3.2 What follows without further invention

Every capability listed below is a consequence of over-determination rather than an additional subsystem. This is the reason the architecture stays small.

- **Virtual sensing.** If a quantity is recoverable along several paths, any one path can supply it when its sensor is absent or has failed. Unmeasured quantities — peak cylinder pressure, knock margin, turbocharger shaft speed, per-cylinder combustion efficiency — are recovered the same way. This is analytical redundancy, and it is what DRDO means by virtual sensing.
- **Sensor-fault discrimination.** A transducer appears in only some of the consistency relations. A component appears in others. The two therefore excite different residual subsets.
- **Fault isolation without hand-authored signatures.** The set of relations a given fault violates is derivable from the equations, not asserted from experience.
- **Validation without ground truth.** If the redundant estimates agree across the operating envelope, the model reproducing them is correct in the quantities that matter — and no run-to-failure dataset was required to establish it.
- **Cross-engine transfer.** Expressed in dimensionless similarity variables, the consistency relations are properties of the engine *class* rather than the individual machine (§7).

3.3 An honest limit on the claim

Parity residuals are still compared against a decision threshold; over-determination changes *what* is thresholded, not the existence of a threshold. The advantage is that the thresholded quantity is an operating-point-invariant consistency error with a near-zero nominal value and a characterisable noise distribution, rather than a raw signal whose acceptable range depends on flight condition. Statistical decision theory then applies cleanly, which it does not to raw channels.

4 Physics of the Engine Twin

4.1 Choice of fidelity

The twin uses a **mean-value engine model** (MVEM): cycle-averaged states, no resolution of individual crank angles. This is the correct fidelity for health monitoring. A crank-angle-resolved model would add in-cylinder pressure detail that no flight sensor observes, at a computational cost that precludes both real-time operation and the forward Monte Carlo propagation required by the mission layer (§11). The MVEM runs several hundred times faster than real time on a single core, which is what makes the reliability computation affordable.

4.2 State equations

The engine is represented by five state groups.

INTAKE MANIFOLD — filling and emptying

$$\frac{dp_{im}}{dt} = \frac{RT_{im}}{V_{im}} (\dot{m}_c - \dot{m}_a) \quad (1)$$

where $\dot{m}_c$ is compressor delivery and $\dot{m}_a$ the flow ingested by the cylinders. Turbocharger degradation manifests here first, as an inability to hold commanded boost.

CYLINDER INDUCTION — speed-density, four-stroke

$$\dot{m}_a = \eta_v(p_{im}, N) \frac{p_{im}}{RT_{im}} \frac{V_d N}{120} \quad (2)$$

with N in rpm and V_d the swept volume; the factor 120 is two crank revolutions per cycle multiplied by 60 s/min. Volumetric efficiency η_v is the channel through which ring wear, valve-seat recession and induction restriction enter the model.

CRANKSHAFT DYNAMICS

$$J \frac{d\omega}{dt} = T_{ind} - T_{fric} - T_{pump} - T_{load}, \quad T_{ind} = \frac{\eta_i \dot{m}_f Q_{LHV}}{\omega} \quad (3)$$

Indicated torque is accumulated per cylinder, so suppressing one cylinder's contribution produces a physically correct misfire signature rather than an imposed one. For a four-stroke engine each cylinder fires once per 720° of crank rotation, so a single-cylinder defect appears at **0.5 engine order** in the angular-velocity spectrum — the classical misfire discriminant, recoverable from the existing crank position sensor.

CYLINDER HEAD THERMAL STATE — one per cylinder

$$m c_p \frac{dT_{cht,i}}{dt} = \dot{Q}_{gas,i} - h_{air} A_{fin} (T_{cht,i} - T_{cool}) \quad (4)$$

Four independent thermal states are what make per-cylinder localisation physically meaningful. Gas-side heat transfer follows a Woschni-type correlation, $h_c = 3.26 B^{-0.2} p^{0.8} T^{-0.55} w^{0.8}$, with B the bore and w a characteristic gas velocity.

TURBOCHARGER SHAFT

$$J_{tc} \omega_{tc} \frac{d\omega_{tc}}{dt} = \eta_m P_{turb} - P_{comp}, \quad P_{comp} = \frac{\dot{m}_a c_p T_1}{\eta_c} \left[\left(\frac{p_2}{p_1} \right)^{\frac{\gamma-1}{\gamma}} - 1 \right] \quad (5)$$

Scaling η_c downward reproduces compressor degradation, which by construction is invisible at sea level and becomes mission-limiting above critical altitude.

4.3 Atmosphere

The ISA troposphere supplies the environmental scenarios the problem statement requires:

$$T_{\infty} = 288.15 \text{ K} - 0.0065 \text{ K/m} \cdot h, \quad p_{\infty} = 101325 \text{ Pa} (1 - 2.25577 \times 10^{-5} h)^{5.2559} \quad (6)$$

An additive offset gives ISA+20 hot-and-high conditions representative of the Leh trials. Above the engine's critical altitude — 11 000 ft for the VRDE unit — the turbocharger can no longer hold rated manifold pressure and power lapses; the twin reproduces this without special-casing.

4.4 Degradation dynamics and ground-truth labels

Wear accumulates through physically motivated laws rather than injected noise. The model carries an explicit scalar damage state $D \in [0, 1]$ per component:

$$\text{Adhesive/abrasive wear (Archard): } \frac{dD}{dt} \propto \frac{k F_n v}{H} \quad (7)$$

$$\text{Thermally activated ageing (Arrhenius): } \frac{dD}{dt} \propto \exp\left(-\frac{E_a}{R T_{cht}}\right) \quad (8)$$

$$\text{Cumulative damage (Miner): } D = \sum_i \frac{n_i}{N_i}, \quad \text{failure at } D = 1 \quad (9)$$

and the corresponding remaining useful life follows directly:

$$\text{RUL} = \frac{1 - D}{dD/dt} \quad (10)$$

Because D is a state of the simulator, every generated sample carries an **exact** RUL label with no censoring. This is the same construction NASA used to produce the C-MAPSS turbofan degradation datasets, which have served as the field's benchmark since 2008. The synthetic origin of the training data is therefore a methodological choice with precedent, not a limitation to be apologised for — provided, as §13 sets out, the model is independently validated.

5 Parity Space: Redundant Estimation

5.1 Four independent estimates of air mass flow

Air mass flow is the single most informative quantity in the engine and is almost never instrumented directly. It is recoverable along four paths that share no common failure mode.

PATH 1 — SPEED-DENSITY From manifold pressure, manifold temperature and crankshaft speed, via Eq. (2):

$$\dot{m}_a^{SD} = \eta_v \frac{p_{im}}{R T_{im}} \frac{V_d N}{120}$$

Sensors: MAP, IAT, crank speed. Model dependency: volumetric efficiency map.

PATH 2 — COMPRESSOR OPERATING POINT From turbocharger shaft speed and pressure ratio, read against the corrected compressor map:

$$\dot{m}_a^C = \dot{m}_{corr} \left(\pi_c, \frac{N_{tc}}{\sqrt{\theta_1}} \right) \cdot \frac{\delta_1}{\sqrt{\theta_1}}, \quad \pi_c = \frac{p_2}{p_1}, \quad \theta_1 = \frac{T_1}{T_{ref}}, \quad \delta_1 = \frac{p_1}{p_{ref}}$$

Sensors: turbo speed, compressor inlet and outlet pressure and temperature. Model dependency: compressor map.

PATH 3 — FUEL AND EXCESS-AIR RATIO From commanded fuel delivery and the measured exhaust oxygen content:

$$\dot{m}_a^\lambda = \lambda \cdot \text{AFR}_{st} \cdot \dot{m}_f$$

Sensors: injector command (or fuel flow meter), wideband lambda / UEGO. Model dependency: none of consequence.

PATH 4 — INTAKE RESTRICTION From a metering restriction, using the compressible orifice relation:

$$\dot{m}_a^R = C_d A \frac{p_{us}}{\sqrt{R T_{us}}} \Psi \left(\frac{p_{ds}}{p_{us}} \right), \quad \Psi(r) = \begin{cases} \sqrt{\frac{2\gamma}{\gamma-1} \left(r^{2/\gamma} - r^{\frac{\gamma+1}{\gamma}} \right)} & r > r_{crit} \\ \sqrt{\gamma} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{2(\gamma-1)}} & r \leq r_{crit} \end{cases}$$

Sensor-set dependency, stated honestly. Path 4 requires either a throttle body or a metering restriction. A FADEC-controlled aero-diesel is unthrottled, so on that installation Path 4 is available only if a hot-film MAF or venturi is fitted. The framework degrades gracefully: with n available paths it produces $n-1$ independent consistency relations, and the isolability analysis in §6 is recomputed for whichever sensor set the target engine carries.

5.2 Independent parity residuals

With n estimates of one quantity, only $n-1$ residuals are linearly independent — the third pairwise difference is the sum of the first two. Choosing Path 1 as reference:

$$\rho_1 = \dot{m}_a^{SD} - \dot{m}_a^C, \quad \rho_2 = \dot{m}_a^{SD} - \dot{m}_a^\lambda, \quad \rho_3 = \dot{m}_a^{SD} - \dot{m}_a^R \quad (11)$$

Stating this explicitly matters: a system claiming six residuals from four estimates is double-counting, and an evaluator familiar with parity-space methods will check.

5.3 Energy closure

The first law supplies a further, entirely independent relation. Chemical power in must equal useful work out plus every loss path:

$$\rho_4 = \dot{m}_f Q_{LHV} - \underbrace{\frac{2\pi N}{60} T_b}_{\text{brake}} - \underbrace{(\dot{m}_a + \dot{m}_f) c_{p,ex} (T_{egt} - T_\infty)}_{\text{exhaust enthalpy}} - \underbrace{\sum_i hA(T_{cht,i} - T_{cool})}_{\text{heat rejection}} \quad (12)$$

In a correct model $\rho_4/\dot{m}_f Q_{LHV}$ sits within a few per cent across the envelope. It is sensitive to any fault that changes where the fuel's energy goes — which is most of them — and is therefore a good detector but a poor isolator on its own.

5.4 Power closure without a torque sensor

Flight engines are not instrumented for shaft torque, but a propeller is itself a calibrated dynamometer. Absorbed power follows the propeller power coefficient:

$$P_{prop} = C_P(J) \rho n^3 D^5, \quad J = \frac{V_\infty}{nD} \quad (13)$$

with n in revolutions per second. Comparing this against the power implied by the fuel path gives a fifth independent relation:

$$\rho_5 = (\eta_i \dot{m}_f Q_{LHV} - P_{fric} - P_{pump}) - C_P(J) \rho n^3 D^5 \quad (14)$$

ρ_5 is the primary channel for friction-related degradation — bearing wear, lubrication breakdown — because those faults consume shaft power without altering the gas path.

5.5 Local relations

Four further relation groups complete the set: per-cylinder exhaust and head temperature deviation from the multi-cylinder conditional mean ($\rho_6 \dots \rho_9$); the oil pressure model residual as a function of speed and oil temperature (ρ_{10}); and the magnitude of the 0.5-order component of crank angular velocity (ρ_{11}), which is zero for a balanced engine and non-zero for any single-cylinder defect.

5.6 The residual vector

The complete parity vector $\rho \in \mathbb{R}^{11}$ is the system's only input to every downstream model. It has three properties that raw telemetry does not: its nominal value is zero, it is invariant to operating point by construction, and its noise distribution can be characterised once on healthy data and reused everywhere. This is why the anomaly detector in §9 does not need to relearn the flight envelope, and why it does not raise an alarm every time the throttle moves.

6 Fault Isolation from Structured Residuals

6.1 Derivation rather than assertion

Because each parity relation is built from a known subset of sensors and a known piece of physics, the effect of any fault on any relation is derivable. Ring wear reduces true volumetric efficiency while the model retains its nominal value, so Path 1 over-reads and the other three remain correct. Compressor fouling degrades the machine while the map stays nominal, so Path 2 over-reads alone. Injector fouling makes delivered fuel fall below commanded fuel; the mixture leans, measured λ rises, and Path 3 over-reads by exactly the ratio of commanded to delivered fuel. A MAP transducer reading high corrupts Path 1 upward *and* Path 2 downward simultaneously, because the same pressure enters one as a density and the other as a pressure ratio — a correlated double effect no component fault can imitate.

THE ISOLATION PRINCIPLE

Every fault has an **odd path**: the estimate that departs while the others hold. Identifying which path is the outlier, and with what sign, identifies the fault — without a classifier, and without ever having observed that fault before.

6.2 Fault incidence matrix

FAULT MODE	ρ_1	ρ_2	ρ_3	ρ_4	ρ_5	ρ_{6-9}	ρ_{10}	ρ_{11}
Ring wear / blow-by	↑	↑	↑	↑	↑	·	↓	·
Compressor fouling	↓	·	·	↑	·	·	·	·
Injector fouling, cyl i	·	↓	·	↑	·	↑	·	↑
Fuel filter / rail loss	·	↓	·	↑	↓	·	·	·
Ignition misfire, cyl i	·	·	·	↑	↓	↓	·	↑
Cooling degradation	·	·	·	↑	·	↑	↑	·
Oil leak / pump wear	·	·	·	·	↑	·	↓	·
Bearing wear	·	·	·	↑	↑	·	↓	·
Detonation / pre-ignition	·	·	·	↑	·	↑	·	·
MAP sensor drift	↑	↑	↑	↑	·	·	·	·
EGT sensor drift, cyl i	·	·	·	↑	·	↑	·	·
Lambda sensor drift	·	↑	·	·	·	·	·	·

↑ strong positive excitation ↑ weak positive ↓↓ negative · unexcited. Component faults above the rule, instrumentation faults below it.

6.3 Isolability analysis

Two faults are structurally isolable if and only if their columns in the incidence matrix differ. Applying that test honestly:

- **Strongly isolable from steady-state parity alone (7 modes).** Compressor fouling, oil system faults, bearing wear, cooling degradation, ring wear, fuel rail loss and lambda sensor drift each produce a unique excitation pattern.
- **Isolable only with dynamic channels.** Injector fouling and ignition misfire on the same cylinder share their steady-state signature and separate on ρ_4 magnitude and on the harmonic content of ρ_{11} — misfire removes a complete combustion event, injector fouling attenuates it.

- **Weakly isolable, requiring active resolution.** *Injector fouling on cylinder i against EGT sensor drift on cylinder i* differ structurally only through ρ_2 and ρ_{11} , both of which are small at low fault severity — exactly the regime where early detection is valuable. *Ring wear against MAP drift* differ only in the relative magnitude of ρ_1 versus ρ_2 , not in sign pattern.

The final category is not a weakness to be concealed; it is the precise, derived justification for the active diagnosis layer in §10. The system knows *which* ambiguities its own structure cannot resolve, and probes only those.

7 Dimensionless Invariants and Cross-Engine Transfer

7.1 Similarity variables for reciprocating engines

Turbomachinery has used corrected mass flow and corrected speed for seventy years precisely so that performance maps collapse across inlet conditions. The same reasoning applies to a reciprocating engine. Expressing the health state in similarity variables rather than absolute readings removes altitude, ambient temperature and — to first order — engine size:

VARIABLE	DEFINITION	WHAT IT ISOLATES
Volumetric efficiency	$\eta_v = \frac{120 \dot{m}_a RT_{im}}{p_{im} V_d N}$	Induction and sealing health
Equivalence ratio	$\varphi = 1/\lambda$	Mixture preparation, injector condition
Brake thermal efficiency	$\eta_{bth} = \frac{P_b}{\dot{m}_f Q_{LHV}}$	Overall conversion quality
Normalised heat rejection	$\frac{\dot{Q}_{cool}}{\dot{m}_f Q_{LHV}}$	Cooling system and combustion phasing
Corrected speed	$N/\sqrt{\theta_{im}}$	Operating point, altitude-independent
Load fraction	BMEP/p_{im}	Effectiveness of delivered charge
Mean piston speed	$\bar{S}_p = 2LN/60$	Mechanical duty, comparable across sizes

7.2 The health manifold

In this space a healthy engine of a given class does not occupy a volume; it occupies a low-dimensional **surface**. Every legitimate operating point — sea level takeoff, 25 000 ft cruise, hot-and-high loiter — maps onto that same surface. Degradation is displacement off it, and the direction of displacement is the fault signature.

This is what makes the anomaly problem tractable with modest data. The detector is not learning a four-dimensional flight envelope; it is learning one surface and measuring distance from it.

7.3 Transfer, and the honest version of the claim

Because the manifold is defined in similarity variables, it is a property of the engine class. A model fitted on one engine should apply to another of the same architecture without retraining. The demonstration is direct: fit on the 180 hp aero-diesel profile, run unmodified on a Rotax 914-class profile, and show the residuals remain centred.

The invariance is first-order, not exact. Bore-to-stroke ratio, combustion system, cylinder count and cooling architecture shift the surface by a small bias. The defensible claim is therefore: **transfer requires on the order of ten healthy operating points from the new engine to remove the bias, rather than a full run-to-failure campaign**. That is a weaker statement than “it transfers perfectly” and a far stronger one than anything a model trained on absolute readings can offer, because a run-to-failure campaign on an aero engine costs years.

8 Platform Structure That Nobody Exploits

8.1 The second engine is a free reference channel

A MALE UAV of this class is twin-engined. Two nominally identical engines, built to the same specification, drawing from the same fuel tanks, flying the same profile through the same air, for eighteen hours. This is the best controlled experiment available in aviation and it costs nothing to use.

Define the cross-engine differential on every channel:

$$\Delta_k(t) = y_k^{(A)}(t) - y_k^{(B)}(t) \quad (15)$$

Common-mode variation — ambient temperature, fuel batch quality, air density, altitude — appears identically on both engines and cancels exactly. What survives the subtraction is engine-specific, which is to say, it is degradation or instrumentation. The differential channel has a noise floor far below either absolute channel, so slow drifts become visible earlier.

The logic is worth stating as an operational rule, because it is how a maintenance officer already thinks: *both engines drifting together is the environment or the fuel; one engine drifting alone is that engine*. Every team that models a single engine forfeits this entirely.

8.2 The loiter orbit is a repeated calibration point

Endurance missions are not a continuous sweep through the operating envelope. They are dominated by hours of racetrack loiter at a fixed altitude and a fixed power setting. Each circuit returns the engine to an operating point that is, to a very close approximation, identical to the previous circuit's.

That is same-point sampling, and it is the cleanest possible basis for trend estimation. Conventional degradation trending must regress across a varying envelope and absorb the resulting operating-point variance into its uncertainty. Here that variance is simply absent: consecutive laps are directly comparable, and the degradation trend is recovered from the difference between like and like.

Detection sensitivity improves as the square root of the number of comparable samples. An 18 h endurance mission at a 20 min orbit period supplies roughly 54 matched samples per flight, at zero instrumentation cost, purely by recognising what the aircraft is already doing.

8.3 Operation above critical altitude

The VRDE engine holds rated power to 11 000 ft; the platform operates to 28 000 ft. Above critical altitude the turbocharger is the binding constraint on available power, and compressor condition — which Eq. (5) shows enters directly — becomes the dominant determinant of whether the aircraft can hold altitude. A degradation that is entirely invisible in a sea-level ground run is mission-limiting at operating altitude. The twin, being physics-based, extrapolates this correctly; a model fitted to absolute readings at low level does not.

9 Estimation and Learning Layer

9.1 Joint state–parameter estimation: a twin that re-tunes itself

A twin calibrated once and never revised is wrong within a few hundred hours, and when it disagrees with the engine there is no way to know which of the two has moved. PRAMANA resolves this by placing the degradation parameters inside the estimator’s state vector:

$$\mathbf{x}_{aug} = \begin{bmatrix} \mathbf{x} \\ \theta \end{bmatrix}, \quad \theta = [\eta_v^{sc}, \eta_c^{sc}, (hA)^{sc}, C_{d,inj}^{(1..4)}, f_{fric}] \quad (16)$$

with the parameters modelled as a slow random walk, $\theta_{k+1} = \theta_k + \mathbf{w}_k$, $\mathbf{w}_k \sim \mathcal{N}(0, \mathbf{Q}_\theta)$ and $\mathbf{Q}_\theta$ small. An unscented Kalman filter then tracks both simultaneously.

Two consequences follow. First, the twin stays synchronised with a physically ageing engine without manual recalibration. Second — and this is the point — **the parameter estimates are themselves the health indicators**, and they are physically interpretable. “Compressor efficiency scale 0.87, down from 0.94, standard deviation 0.02” is a statement a propulsion engineer can accept or dispute. “Health score 0.71” is not.

The filter also supplies the rigorous form of sensor-fault discrimination. The innovation $\nu_k = \mathbf{y}_k - h(\hat{\mathbf{x}}_{k|k-1})$ measures surprise. A component fault produces innovation that is *absorbed* by an adjustment of θ ; a transducer fault produces innovation on one channel that no physically admissible parameter change can explain. Monitoring the normalised innovation squared alongside the parameter trajectory separates the two cases on principle rather than by heuristic.

9.2 Physics-guided residual estimator

Where a learned component is needed, it corrects the physics rather than replacing it:

$$\hat{y} = \underbrace{f_{phys}(u; \theta)}_{\text{MVEM}} + \underbrace{g_{NN}(u)}_{\text{small correction}} \quad (17)$$

The network learns only the discrepancy, so it trains on far less data and cannot produce physically absurd predictions at operating points outside its training set — which matters, because the training data will be dominated by mid-altitude cruise and the aircraft will fly to 28 000 ft. Soft physical constraints enter the loss:

$$\mathcal{L} = \mathcal{L}_{data} + \lambda_1 \underbrace{\|\rho_4\|^2}_{\text{energy closure}} + \lambda_2 \underbrace{\|\dot{m}_a + \dot{m}_f - \dot{m}_{ex}\|^2}_{\text{mass continuity}} + \lambda_3 \underbrace{\left\| \min\left(0, \frac{\partial T_{cht}}{\partial \dot{m}_f}\right) \right\|^2}_{\text{monotonicity}} \quad (18)$$

This is a physics-guided hybrid model, not a PINN in the strict sense of enforcing PDE residuals by collocation. That distinction should be stated rather than blurred: the strict formulation offers little additional value for a lumped-parameter ODE system, and carries real convergence risk. What is claimed — a physics backbone with a learned correction and physically motivated loss terms — is what is delivered.

9.3 Anomaly detection, classification and remaining useful life

An LSTM autoencoder is trained on windows of the *parity vector* ρ from healthy operation. Because ρ is nominally zero and operating-point invariant, the autoencoder learns a genuinely simple target and does not react to throttle or altitude changes — the failure mode that dominates

naive implementations trained on raw telemetry. The decision threshold is the 99.5th percentile of reconstruction error on held-out healthy data, combined with an N -of- M persistence rule to suppress transients.

Fault classification runs a temporal convolutional network over the same window, and is reported *alongside* the incidence-matrix match of §6. The network supplies accuracy; the incidence match supplies the explanation, and agreement between two independent mechanisms is itself evidence.

Remaining useful life is estimated by two heads. A quantile-regression network returns $p_{10}/p_{50}/p_{90}$ directly, and a physics head integrates the identified damage rate forward under the planned mission. The conservative of the two is reported, and disagreement between them exceeding the predictive interval is surfaced as a warning in its own right.

10 Active Diagnosis

10.1 Resolving what structure cannot

Section 6 identified the ambiguities that steady-state parity cannot break at low fault severity. The most important is injector fouling on cylinder i against a drifting exhaust gas temperature sensor on cylinder i : both raise the indicated $T_{egt,i}$ relative to its siblings, and at small amplitude the corroborating channels are below the noise floor.

Rather than guess, the system **asks the engine a question**. It commands a small, bounded perturbation of cylinder i fuel trim and examines the *gain* of the response rather than its level.

10.2 Why gain discriminates and level does not

Let the commanded trim perturbation be $u(t)$, small and zero-mean. Under the injector hypothesis H_1 , the injector delivers a reduced fraction $\kappa < 1$ of what is commanded, so the physical fuel perturbation is $\kappa u(t)$ and the exhaust temperature responds with attenuated amplitude. Under the sensor hypothesis H_2 the injector is healthy, the physical response is full, and the transducer merely adds a constant offset b — which does not appear in the alternating component at all. Estimating the gain by cross-correlation,

$$\hat{G} = \frac{\text{cov}(y_{egt,i}, u)}{\text{var}(u)} \quad \Rightarrow \quad \hat{G}|_{H_1} \approx \kappa G_0, \quad \hat{G}|_{H_2} \approx G_0 \quad (19)$$

The bias b cancels identically. A sequential probability ratio test on the log-likelihood ratio $\Lambda = \log[p(y | H_1)/p(y | H_2)]$ continues the probe only until a pre-specified error bound is met, typically within a few seconds.

The probe itself is chosen to maximise separation subject to authority and safety limits:

$$u^* = \arg \max_{\|u\| \leq u_{max}} \left\| \mathbb{E}[y | H_1, u] - \mathbb{E}[y | H_2, u] \right\|_{\Sigma^{-1}}^2 \quad (20)$$

10.3 Safety envelope

Active perturbation of a flying engine requires an explicit safety case, and the following constraints form it. Probe amplitude remains inside the FADEC's existing trim authority ($\pm 3\%$ to $\pm 5\%$), which the controller already exercises during normal closed-loop operation. Duration is bounded to a few seconds. The probe is armed only when posterior fault probability lies in the ambiguous band, so it is never routine. It is inhibited during takeoff, climb, approach and any single-engine condition, and aborted immediately if any monitored parameter approaches a certified limit. On a twin-engine aircraft it is further restricted to one engine at a time.

This capability changes the system's category. A monitor observes and infers. PRAMANA, when its own structure tells it that observation is insufficient, designs and executes an experiment.

11 Mission Reliability Layer

11.1 From diagnosis to decision

The problem statement asks for mission reliability *enhancement*. Enhancement requires acting on the aircraft's operation, not annotating it. Because the twin is a fast forward model and the estimator supplies a posterior over the degradation state, the natural output is a probability.

$$P(\text{complete}) = \frac{1}{M} \sum_{j=1}^M \mathbb{I} \left[\text{twin}(\theta_j, \text{route}_{\text{remaining}}) \text{ stays within limits} \right], \quad \theta_j \sim \mathcal{N}(\hat{\theta}, \Sigma_{\theta}) \quad (21)$$

$M = 200$ samples is sufficient and completes in well under a second, because the mean-value model runs far faster than real time. This is the return on choosing the right modelling fidelity in §4.

11.2 Life-extending control

Given that machinery, the system can do more than report the probability — it can search for the operating schedule that improves it. Damage accumulation depends strongly on power setting and cylinder head temperature, so trading a small amount of performance buys a disproportionate amount of life:

$$u^* = \arg \max_{u \in \mathcal{U}} V_{\text{mission}}(u) \quad \text{subject to} \quad P(\text{complete} \mid u) \geq 1 - \alpha \quad (22)$$

where u is the cruise power and boost schedule, V_{mission} is time on station, and α is the accepted risk. The output is a renegotiated operating envelope handed to the flight management system, not an alarm handed to an operator.

11.3 What the operator actually sees

OUTPUT	FORM	WHY IT IS ACTIONABLE
Completion probability	Continue 62 % Derate 94 % RTB 99 %	Three options with quantified risk, not a red light
Cost of the safe option	"Derating costs 40 min on station"	Reliability advice that ignores mission value is ignored advice
Point of no return	Moving marker on the route	Computed from fuel remaining and <i>degraded</i> BSFC, not the book figure
Recommended envelope	Power and boost ceiling	Directly consumable by the flight management system

12 System Architecture

12.1 Why the edge/cloud split is arithmetic, not fashion

Knock and bearing signatures occupy the 5 kHz to 20 kHz band. A MALE UAV's datalink to its ground station carries tens of kilobits per second, shared with command traffic and sensor video. Raw vibration cannot be downlinked, in any architecture, ever. The high-rate signal processing must therefore execute on the airframe, and only extracted features and health state travel down the link. The problem statement's requirement for both edge computing and cloud analytics is a consequence of this bandwidth arithmetic, and stating it that way is more persuasive than any architectural preference.

Layer 0 · Engine and instrumentation

ECU on CAN · CANaerospace / DroneCAN framing · in simulation, the MVEM of §4

Layer 1 · Two-lane telemetry

fast lane 5–20 kHz on-board only · slow lane 10–50 Hz to ground · health vector 1 Hz

Layer 2 · Edge (on the airframe)

parity residual generation · UKF joint state–parameter estimation · LSTM-AE anomaly

Layer 3 · Ground analytics

incidence-matrix isolation · fault classifier · RUL with p10/p50/p90 · fleet history

Layer 4 · Decision

Monte Carlo mission propagation · P(complete) · life-extending control · derate advisory

Layer 5 · Ground control station

live twin overlay · 3D engine with per-cylinder state · replay scrubber · post-flight report

Layers 0 and 1 are the only components that know the data is simulated. Replacing them with a physical ECU and a real datalink changes nothing above them — which is the entire deployment argument, and is why it is worth designing the boundary deliberately.

12.2 Telemetry and bus

Engine telemetry is framed as CAN rather than as an application-level message format, because that is what the engine controller actually emits. CANaerospace — developed by Stock Flight Systems in 1998 and adopted by NASA as the AGATE databus standard in 2001, reserving identifiers 300 to 1799 — and DroneCAN, which ArduPilot and PX4 run at 1 Mbit/s, are the two relevant conventions. In development the bus is a Linux virtual CAN interface, so the ground software consumes byte-identical frames whether the source is the simulator or an engine.

13 Validation Without Ground Truth

No run-to-failure dataset exists for UAV aero piston engines. This is not an oversight in the present work — it is the reason DRDO has issued a request for proposals in this area rather than buying a product. Any credible submission must therefore answer the validation question directly instead of deferring it.

13.1 Level 1 — internal consistency

The over-determined structure of §5 validates the twin against itself. If three or four independent estimates of air mass flow agree across the full operating envelope, then the model reproduces the induction, boost and combustion physics correctly *in the quantities that matter*, and no external reference was required to establish it. The reported metric is the distribution of each parity residual, normalised, over the envelope. This is a real answer to the hardest question the panel can ask.

13.2 Level 2 — cross-engine consistency

The same models, applied unmodified to a second engine profile with different displacement and rating, must produce centred residuals after a small bias correction (§7). This tests that the system learned physics and not the idiosyncrasies of one simulator.

13.3 Level 3 — external benchmark

The residual $\rightarrow$ autoencoder $\rightarrow$ RUL pipeline is run unmodified on NASA's turbofan degradation datasets from the Prognostics Center of Excellence repository — C-MAPSS, and preferably N-CMAPSS, which carries real flight conditions. A competitive score on data the team did not generate is the only conclusive evidence that the learning stack is not memorising its own simulator.

13.4 Reported metrics

METRIC	DEFINITION	WHY IT IS REPORTED
Detection lead time	Interval between twin alert and limit exceedance	The operational value proposition
False alarms per flight hour	Spurious alerts over healthy flights	The question a maintenance officer asks first
NASA asymmetric score	$\sum e^{-d/13} - 1$ if $d < 0$; $\sum e^{d/10} - 1$ otherwise	Field-standard; penalises late prediction harder
Prognostic horizon, α - λ	Standard PHM prognostic measures	Demonstrates familiarity with the discipline
Parity residual spread	Normalised, across envelope	Level 1 validation evidence
Isolation confusion matrix	Across all fault modes	Honest reporting of what does not separate

14 The DRDO Perspective

14.1 Mapping to the expected solution components

§	REQUIREMENT	DELIVERED BY
A	Digital twin core, synchronised, modular	Mean-value engine model (§4) driven by live telemetry, with engine geometry and maps held in a configuration profile so a new engine is a data change
B	Health monitoring across the full channel set	Parity relations (§5) plus per-cylinder thermal states; battery, alternator and injection timing carried as monitored channels
C	Fault detection incl. misfire, injector, lubrication, <i>sensor drift</i>	Incidence-matrix isolation (§6) with instrumentation faults treated as first-class fault modes, resolved actively where structurally ambiguous (§10)
D	AI/ML: anomaly, RUL, trend, advisory	UKF joint estimation, LSTM autoencoder on parity residuals, two-headed RUL with predictive intervals (§9)
E	Simulation and replay, environment, endurance	ISA model with hot-and-high offset, high-altitude and 18 h endurance presets, full-state mission replay
F	Visualisation dashboard	Ground control station with live twin overlay, per-cylinder 3D state, replay scrubber and post-flight report

14.2 Alignment with DRDO's own programme

DRDO has issued a request for proposals under the Technology Development Fund for a digital twin framework for aero engine health monitoring, integrated with Health and Usage Monitoring Systems. That programme is structured in two phases: first, a digital model of the engine built from physics-based *and* data-driven approaches covering mechanical behaviour, thermal behaviour, degradation pathways and failure modes; second, integration into the wider aircraft digital ecosystem.

PRAMANA is scoped as a Phase 1 contribution, on the reciprocating rather than the gas-turbine side of the portfolio. The alignment is worth stating plainly in evaluation: this is not a student exercise adjacent to an organisational interest, it is a prototype of a capability the organisation has formally solicited.

14.3 Indigenous powerplant context

The natural target is the VRDE 180 hp aero-diesel: DRDO-developed, rated to 11 000 ft, running an indigenously developed FADEC, with over 1100 hours of test-profile running and high-altitude trials at Leh and Changla to 17 664 ft. A FADEC is precisely the interface a twin requires — it already computes and publishes the commanded quantities against which delivered quantities are compared.

Health monitoring is also a direct contributor to the indigenisation objective. A domestically developed engine accumulates operational credibility through demonstrated reliability, and a twin that converts every flight hour into quantified component-level evidence accelerates that far faster than fixed-interval overhaul does.

14.4 Technology readiness and deployment roadmap

PHASE	TRL	ACTIVITY	EVIDENCE PRODUCED
Current	3 → 4	Software twin, synthetic generation, internal consistency and external benchmark validation	Parity spread, N-CMAPSS score, isolation confusion matrix
Next	4 → 5	Recalibration against instrumented dynamometer data from a physical engine; seeded-fault testing	Detection lead time and false alarm rate on real hardware
Then	5 → 6	Edge deployment on an airborne computing module; flight-test integration; recorded-data shadow mode	Latency, resource envelope, agreement with maintenance findings
Mature	6 → 7	Fleet analytics; maintenance-record feedback closing the diagnostic loop	Continuously improving priors from confirmed outcomes

The final row identifies the gap that every deployed prognostic system in service today still suffers from: a diagnosis is issued, the aircraft lands, maintenance opens the component, and the finding never returns to the algorithm. Designing that feedback path in from the beginning is inexpensive now and effectively impossible to retrofit later.

15 Why This Approach Is Strong

15.1 Against the alternatives

APPROACH	FAILS BECAUSE	PRAMANA
Threshold monitoring	Late by construction; blind to context; cannot doubt its own sensors	Compares measurements to each other; operating-point invariant; instrumentation faults are a modelled class
Learned model on raw telemetry	Learns the flight envelope; false-alarms on every transient; cannot extrapolate above training altitude	Learns one invariant surface; physics backbone extrapolates; corrections stay bounded
Physics twin with fixed parameters	Diverges from an ageing engine; disagreement is uninterpretable	Parameters are estimated states; the estimates <i>are</i> the health indicators, in physical units
Detection-only systems	Answers “what is wrong”, never “what should I do”	Returns a completion probability and a recommended operating envelope

15.2 The five questions this design answers that others cannot

1. **How do you know your twin is right, with no real data?** Because it is over-determined and validates against itself. Independent estimates of the same quantity agree, or they do not, and no external reference is needed to tell which.
2. **Is it the engine or the sensor?** A component fault is absorbed by a physically admissible parameter change; an instrumentation fault is not. Where the structure genuinely cannot separate the two, the system says so and runs an experiment.
3. **Does it transfer to a different engine?** The health state is expressed in similarity variables, so the manifold belongs to the engine class. Transfer costs on the order of ten healthy samples, not a run-to-failure campaign.
4. **What does the health number mean physically?** Compressor efficiency scale, volumetric efficiency scale, injector discharge coefficient — quantities a propulsion engineer can argue with, each with a covariance.
5. **What do I do about it?** Continue, derate, or return, each with a probability, and the cost of the safe option stated in minutes on station.

15.3 What is deliberately not claimed

The techniques are established, and the document says so. The strict-PINN formulation is not used, and the reason is given. Three fault pairs are not structurally isolable from steady-state parity, and they are named rather than hidden. Cross-engine invariance is first-order and requires a small bias correction. Parity residuals are still thresholded.

Volunteering these is not modesty. An evaluator who finds a limitation the authors concealed discounts everything else in the document; an evaluator who finds every limitation already identified and bounded reads the remaining claims as reliable. In a technical evaluation that trade is strongly favourable.

References

Problem statement and programme context

Smart India Hackathon 2026, Problem Statement SIH26054, Defence Research and Development Organisation. • DRDO request for proposals, digital twin framework for aero engine health monitoring, Technology Development Fund scheme.

Engine and platform data

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Engine modelling

Heywood, *Internal Combustion Engine Fundamentals* — combustion, heat transfer, volumetric efficiency and performance parameters. • Guzzella and Onder, *Introduction to Modeling and Control of Internal Combustion Engine Systems* — mean-value engine models, filling and emptying, turbocharger dynamics. • Woschni, convective heat transfer correlation for the internal combustion engine.

Fault detection, estimation and prognostics

Gertler, *Fault Detection and Diagnosis in Engineering Systems* — parity relations, structured residuals, fault incidence matrices and isolability. • Kalman-filter-based gas turbine health parameter estimation, NASA Glenn Research Center — joint state and health parameter estimation for aero engines. • Auxiliary signal design for active fault detection — probe design for hypothesis separation. • Life-extending and damage-mitigating control for propulsion systems — trading performance for component life. • Crank angular velocity methods for misfire detection — 0.5 engine order discriminant for four-stroke engines.

Datasets and benchmarks

NASA Prognostics Center of Excellence data repository — *Turbofan Engine Degradation Simulation* (C-MAPSS) and *Turbofan Engine Degradation Simulation-2* (N-CMAPSS). • Saxena, Goebel et al., damage propagation modelling for aircraft engine run-to-failure simulation — the construction underlying C-MAPSS.

Bus and integration standards

CANaerospace, Stock Flight Systems — higher-layer CAN protocol for aeronautical applications, adopted as the NASA AGATE databus standard. • DroneCAN and Cyphal, CAN in Automation — UAV bus conventions used by ArduPilot and PX4.

A note on the name. Pramāṇa (प्रामाण्य) is the Indian epistemological term for a valid means of knowledge — the question of how one establishes that something is true, and how independent sources of knowledge corroborate one another. It was chosen because it describes the system's actual mechanism rather than decorating it. The name is a single macro in the document source and in the codebase, and can be changed in one line.